TRIGGER POINTS (TR.P.) — CANVAS NOTES

ILO's

- To define Tr.Pt.
- To diagnose Tr.Pt.
- To understand Tr.Pt releases Techniques.
- To understand indications & contraindications.
- To integrate the knowledge of Tr.Pt. Management into manual practice.

Epidemiology of Tr.P.

- Age: 30-50 years
- Sex: Female more than male
- Location: postural Ms
- 85% Outpatient clinics

What is trigger points mean?

The inside of the body is covered with soft tissue called fascia. This tissue covers every structure including organs, muscles, nerves and blood vessels. The fascia that covers your muscles is called myofascia. When the myofascia is stressed from overuse or trauma it can tear and adhere together.

These adhesions are called "trigger points" and can prevent the muscles from working well. Trigger points lead to an increase in muscle stiffness and tenderness and a decrease in range-of-motion. In addition, the discomfort from trigger points can radiate from the adhesion. This is called referral pain.

Etiology

- Stress & tension
- Sleep disorder & fatigue
- Neurological factors
- Post-operations
- Chronic micro trauma
- Macro trauma
- Systemic dysfunction
- Ms deficiency

Spasm-Pain-Spasm Cycle

- Muscle spasm
- Splinting phenomenon
- Muscle tension
- Muscle deficiency

Stress Cycle

STRESS

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PAIN

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Muscle Tension

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Pressure on Nerves

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Reduced Circulation

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Muscle Shortening

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Restricted Movement

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Trigger Irritating

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Types of Tr.P.

NOT ALL POINTS ARE TRIGGER POINTS, AND NOT ALL TRIGGER POINTS ARE THE SAME!

TRIGGER POINTS are described according to: Location, Tenderness, Chronicity. As: Central (or primary), Satellite (or secondary), Attachment, Diffuse, Inactive (or latent), Active, Ligamentous trigger points.

MYOFASCIAL TRIGGER POINT

Clinical definition of a central trigger point

A hyperirritable spot in skeletal muscle that is associated with a hypersensitive palpable nodule in a taut band. The spot is painful on compression and can give rise to: Characteristic referred pain, Referred tenderness, Motor dysfunction, Autonomic phenomena.

Etiological definition of a central trigger point

A cluster of electrically active loci each of which is associated with: A contraction knot, A dysfunctional motor endplate in skeletal muscle.

Types of Myofascial Trigger Points

Include: Active, Associated, Attachment, Central, Key, Latent, Primary, Satellite.

(Note especially the distinction between central and attachment myofascial trigger points).

Any myofascial trigger point is to be distinguished from: Cutaneous trigger point, Ligamentous trigger point, Periosteal trigger point, Any other non-muscular trigger point.

CENTRAL (OR PRIMARY) TRIGGER POINTS

These are the most well-established and "florid" points when they are active. Usually what people refer to when they talk about trigger points.

Location

Central trigger points always exist in the center of the muscle belly, where the motor endplate enters the muscle.

Muscle shape and fiber arrangement is of importance in this regard. Example: In multipennate muscles (such as the deltoid): There may be several central points. If muscle fibers run diagonally: This may lead to variations in trigger point location.

SECONDARY TRIGGER POINT (SATELLITE TRIGGER POINTS)

A hyperirritable spot in a muscle or its fascia that became active because its muscle was overloaded as: A synergist substituting for, Or an antagonist countering the tautness of the muscle that contained the primary trigger point.

Trigger points may be "created" as a response to the central trigger point in neighboring muscles that lie within the referred pain zone. The primary trigger point is still the key to therapeutic intervention. The satellite trigger points often resolve once the primary point has been effectively rendered inactive.

As a result: Satellite points may prove resistant to treatment until the primary central focus is weakened. This is often the case in the paraspinal and/or abdominal muscles pain.

ATTACHMENT TRIGGER POINTS

Myofascia is a continuum. The area where the tendon inserts into the bone (tendo-osseous junction) is often tender.

This may be: The result of existing forces travelling across these regions. Or due to an associated chronic, active myofascial trigger point.

The tenderness has been demonstrated to be reduced once the primary central trigger point has been treated. In such cases, the point is described as an attachment trigger point.

If a chronic situation occurs where the primary and attachment trigger points remain untreated: Degenerative changes within the joint may be precipitated and accelerated.

DIFFUSE TRIGGER POINTS

Trigger points can sometimes occur where: Multiple satellite trigger points exist, Secondary to multiple central trigger points. Commonly seen in Severe postural deformity such as: Scoliosis where an entire quadrant of the body is involved.

In this scenario: The secondary points are said to be diffuse. These diffuse trigger points often develop along lines of altered stress and/or strain patterns.

INACTIVE (OR LATENT) TRIGGER POINTS

Applies to lumps and nodules that feel like trigger points. Can develop anywhere in the body. Often secondary.

Characteristics

Not painful. Do not elicit a referred pain pathway.

The presence of inactive trigger points within muscles may lead to: Increased muscular stiffness. It has been suggested that: These points are more common in those who live a sedentary lifestyle.

These points may reactivate if: The central or primary trigger point is re-stimulated. Trauma and injury occur.

ACTIVE TRIGGER POINTS

This can apply to: Central trigger points, Satellite trigger points.

Activation

A variety of stimulants, such as forcing muscular activity through pain, can activate an inactive trigger point. Common when activity is increased after: Road traffic accident (RTA) where multiple and diffuse trigger points may have developed.

Denotes

The trigger point is: Tender to palpation, Elicits a referred pain pattern.

LIGAMENTOUS TRIGGER POINTS

There is evidence that ligaments may develop trigger points. Examples: Sacrotuberous ligament → refer pain to heel. Sacrospinous ligament → refer pain to heel. Iliolumbar ligament → refer pain to groin and even into the testicles or vagina.

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Trigger points in the sacrotuberous ligaments can have profound effects on: Low back pain. Lumbar erector spinae pain, Pelvic pain. May also be associated with: Backache, Neck pain, Vocal dystonia.

Trigger points may manifest in: Anterior longitudinal ligament of the spine (e.g. after whiplash) Result: Neck instability. Fibular collateral ligament: Similar referred pain pattern to vastus lateralis. Ligamentum patellae: Profoundly useful for treating knee pain syndromes.

Eight Differential Criteria and Characteristics for Identifying Myofascial Trigger Points

1. Acute, chronic or persistent overloading of a muscle.
2. Distinct patterns of pain referral inherent to that muscle.
3. Diminished strength and lengthening ability of that muscle.
4. Taut, palpable band or nodule within muscle.
5. Extreme tenderness of palpable band or nodule within fibers.
6. Elicited twitch response when taut band is either stimulated or snapped transversely.
7. Replication of pain pattern when trigger point is stimulated.

8. Asymptomatic relief results when treatment has been performed.

Four Phases of Trigger Points

Phase 1

Active Trigger Point: Palpable button-like nodular mass within muscles, fascia, scar tissue, ligaments, and periosteum. Has characteristics of a trigger point. Produces a subjective referred pain complaint or autonomic phenomena. Present constantly regardless of activity or rest.

Phase 2

Active Trigger Point: Palpable button-like or nodular mass within muscles, fascia, scar tissue, ligaments, and periosteum. Has characteristics of a trigger point. Produces a subjective referred pain complaint or autonomic phenomena during activity. Reduced or eliminated through rest.

Phase 3

Latent Trigger Point: Palpable button-like or nodular mass within muscles, fascia, scar tissue, ligaments, and periosteum. Has characteristics of a trigger point. When snapped transversely it may replicate a referral pattern as if it were active. Without direct pressure, it does not produce a subjective referred pain complaint or autonomic phenomena during either activity or rest.

Phase 4

Muscle tissue does not exhibit a palpable ropy or button-like nodular mass characteristic of a trigger point (TrP). However, the tissue may be: Hypertonic, Indurated, Affecting range of motion, Causing structural deviation, Hypersensitive to tactile pressure.

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